

Requirements Engineering Assignment 2

ParcelResolve

Product Vision, Scope and Stakeholder Analysis

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A. Product Vision

Problem / Need

Every day, large e-commerce and delivery companies process enormous numbers of parcels. Most of them get delivered without any problem. But when something goes wrong — a parcel gets stuck, goes missing, ends up at the wrong address, or the customer disputes the delivery — things can get complicated fast.

The issue is rarely that the organization lacks information altogether. Most large operators already have tracking systems, order management platforms, warehouse tools, and customer service software running in parallel. The data is often somewhere. The problem is that turning that data into a coordinated response is much harder than it sounds.

In practice, resolving a parcel incident often means someone manually chasing updates across different teams, checking multiple systems, and figuring out who is actually responsible for taking the next step. Without a dedicated process to manage this, the same problems tend to come up repeatedly:

- incidents taking longer to resolve than they should;
- the same customer being asked for the same information more than once;
- teams unsure who owns the next action;
- follow-ups being missed or delayed;
- similar cases being handled differently depending on who picks them up;
- no real view of how many unresolved incidents are sitting open at any given time;
- rising operational costs from unnecessary manual work; and
- customers left frustrated by slow or unclear communication.

Stated simply, the core problem is this: how does a delivery organization get from “we know something went wrong with this parcel” to “this incident has been properly resolved” in a way that is fast, consistent, and does not rely on manual coordination at every step?

Intended Users and Customers

ParcelResolve is designed to work as an internal platform inside a large e-commerce or delivery organization — one that already has most of the operational tools it needs, but lacks a dedicated layer for managing what happens when a parcel incident occurs.

The main parties involved are:

- **Host / Customer Organization:** The business that deploys and runs ParcelResolve. It owns the surrounding systems and infrastructure and is ultimately responsible for how incidents are handled.
- **Product Owner:** The person or role that sets the direction for ParcelResolve, decides what gets built and when, and is accountable for whether the product actually delivers value.
- **External Delivery Stakeholders:** The external parties directly involved in or affected by a parcel incident. This includes the sender, recipient, and, where relevant, a third-party logistics or

delivery service provider.

- **Operational User:** Internal staff who use ParcelResolve day to day — customer service agents, warehouse teams, delivery operations, and anyone else who needs to act on or investigate an incident.
- **IT / Integration Team:** The technical team that builds, connects, maintains, and secures ParcelResolve within the organization's existing technical environment.

Main Value

The core value proposition of ParcelResolve is straightforward:

Turn a detected parcel problem into an actionable and trackable resolution process.

This is not meant to be another tracking tool or a generic support ticket system. The idea is that ParcelResolve sits on top of the organization's existing systems and takes over once something goes wrong — pulling in the information that already exists and coordinating what needs to happen next.

An incident can enter ParcelResolve from two main sources:

1. **System-initiated:** An existing operational system flags a potential issue automatically — for example, a parcel that has not moved for an unusually long time.
2. **Customer-initiated:** A customer reports a problem with their delivery. This can happen directly through a customer-facing channel or through a customer service agent who creates the incident on the customer's behalf.

Regardless of how an incident comes in, it goes through the same standardized process. What the system needs to be able to answer at any point is:

- what kind of incident is this?
- what should happen next, and who is responsible for it?
- has the required action actually been completed?
- has the underlying problem actually been fixed?

Main Goals

ParcelResolve should be able to:

1. take in parcel incidents from the supported sources and convert them into a consistent internal format;
2. assess and classify each incident into a supported category so the right workflow can be applied;
3. run the appropriate resolution workflow based on the incident type and the organization's business rules;
4. reduce the need for manual coordination wherever automation can safely handle it;
5. make it clear at all times who is responsible for the next required action;
6. track each incident and its associated actions from start to finish;
7. keep an eye on deadlines, overdue actions, and incidents that have been open too long;
8. escalate cases when predefined thresholds or conditions are not met;
9. handle customer communication appropriately throughout the resolution process;
10. only close an incident once the resolution has actually been verified, not just when an action has been completed; and
11. give the organization visibility into recurring patterns, resolution performance, and operational bottlenecks.

Product Vision Statement

ParcelResolve aims to be a reliable incident-resolution platform that helps large e-commerce and delivery organizations catch parcel problems early, coordinate the right response, and close incidents properly — with as little unnecessary manual effort as possible.

B. Initial Product Scope

Scope Definition

ParcelResolve is an internal parcel incident resolution platform built for large e-commerce and delivery organizations.

Its scope covers everything from the moment a problem is detected or reported, through the investigation and resolution steps, to the point where the incident is verified as resolved and formally closed.

The general flow looks like this:

Detect → Understand → Act → Monitor → Verify Resolution → Close

The exact steps within this flow will vary depending on the incident type and the applicable business rules.

Main Capabilities Within Scope

Incident Initiation

ParcelResolve needs to be able to receive incidents from two main sources:

- incidents flagged automatically by connected operational systems; and
- incidents reported by customers, either directly or through customer-service staff acting on their behalf.

Whatever the source, each incoming incident gets standardized into a common ParcelResolve format before it enters the resolution process. This is what allows incidents to be handled consistently, regardless of how they arrived.

Incident Assessment and Classification

Once an incident is created, ParcelResolve assesses it using available information from the source and from connected systems. The goal is to classify it into one of the supported incident categories:

- delayed parcel;
- potentially lost parcel;
- misdelivered parcel;
- failed delivery; and
- delivery dispute.

Classification drives the rest of the process — the wrong classification early on can send an incident down the wrong path, so this step matters more than it might look.

Incident Information and Status Management

ParcelResolve maintains the working information needed to manage each incident over its lifetime. This includes things like:

- parcel identifier and relevant order reference;
- incident type and current status;
- where the incident came from;
- relevant delivery events pulled from integrated systems;

- who is responsible for what;
- required actions and their deadlines;
- how the incident was resolved; and
- a full history of changes and actions taken.

The intent is to give anyone working on an incident a clear and up-to-date picture of where things stand. This is not intended to turn into a general document archive or an order-management system.

Resolution Workflow Management

ParcelResolve manages predefined resolution workflows that are triggered based on incident type and business rules. These workflows define what steps are needed and what conditions must be met before the incident can move forward toward closure.

Action and Responsibility Management

One of the recurring problems in manual incident handling is that nobody is quite sure whose job it is to do the next thing. ParcelResolve addresses this by creating and tracking specific actions, each with a clearly assigned owner, a priority level, a deadline, and a completion status.

Evidence Capture and Retrieval

For delivery disputes in particular, it is not enough to just track what happened — there needs to be documented evidence to support the resolution decision. ParcelResolve therefore supports the capture and retrieval of relevant evidence, which may include:

- delivery photographs;
- signature records;
- GPS coordinates at the point of delivery;
- system-generated timestamps and event logs; and
- supporting information submitted by operational staff or pulled from integrated systems.

This capability is needed to ensure that dispute outcomes are defensible and consistent. ParcelResolve is not intended to become a primary storage system for this evidence — it links to and surfaces relevant records from where they already exist.

Monitoring and Follow-up

ParcelResolve keeps track of active incidents and their associated actions on an ongoing basis. This means watching for things like:

- deadlines that are getting close;
- actions that have gone overdue;
- incidents that have been open longer than expected;
- new information arriving from connected systems; and
- conditions that suggest the incident may need a different approach, or that resolution might already have been achieved.

Escalation Management

When things are not moving as they should, ParcelResolve escalates. Escalation might mean reassigning the incident to a more senior team, raising the priority, sending a notification, or switching the incident to a different workflow. The specific rules for when and how escalation happens will be defined later as part of the detailed requirements work.

Customer Communication

Customers affected by a parcel incident should not be left in the dark. ParcelResolve supports outbound communication at appropriate points in the process, for example:

- confirming that the incident has been received;
- asking for additional information when needed;

- providing a progress update;
- explaining what happens next; and
- confirming that the issue has been resolved.

The specific channels and the degree of automation will be worked out in later requirements stages.

Resolution Verification and Closure

An important design principle for ParcelResolve is that completing an action is not the same as resolving the incident. For example, if a warehouse locates a missing parcel, the incident should stay open until that parcel has been confirmed back in the delivery process and is on its way to the customer.

Only once the applicable resolution conditions have been verified should the incident be formally closed.

Operational Insights

ParcelResolve should give the organization a useful view of how incident resolution is actually performing. This might include information like:

- how many incidents are open and of what type;
- average time to resolution;
- incidents that are overdue or escalated;
- patterns in recurring problems; and
- resolution outcomes over time.

The exact scope of reporting and analytics will be refined as the project progresses.

Initial Incident Coverage

For the initial version, ParcelResolve will focus on:

- delayed parcels;
- potentially lost parcels;
- misdelivered parcels;
- failed deliveries; and
- delivery disputes.

The final MVP incident set will be confirmed once requirements are further developed.

Integration Boundary

ParcelResolve does not replace the organization's existing operational systems. Instead, it consumes relevant data from them and coordinates the resolution process on top of them.

Systems that ParcelResolve is likely to integrate with include:

- order management systems;
- parcel tracking systems;
- warehouse management systems;
- delivery and driver systems;
- customer service systems; and
- other internal operational platforms as needed.

The specific interfaces and data formats will be defined during later requirements and architecture work.

Out of Scope

The following areas are not part of the initial product scope:

- order creation and order management;
- core parcel tracking infrastructure;
- warehouse management;
- route optimization and driver navigation;
- shipment creation and label generation;
- physical transportation or physical parcel recovery;
- payment processing and compensation workflows;
- damaged-parcel handling;
- wholesale replacement of customer service or operational staff; and
- multi-company or third-party resolution platforms.

ParcelResolve may need to connect to some of these areas, but it does not take over their core functions. In particular, damaged parcels remain outside the initial incident types. If damage is discovered during another type of incident, the case may require human escalation, and the exact handling of such mixed incidents will be addressed in later requirements work.

Product Boundary

ParcelResolve begins when: a parcel incident is detected, reported, or received from a connected system.

ParcelResolve ends when: the incident has been verified as resolved and is formally closed.

Everything in between — assessing the incident, coordinating the response, tracking progress, and confirming resolution — is within scope.

Scope Principle

ParcelResolve is for resolving parcel incidents. It is not trying to replace the organization's order, tracking, warehouse, delivery, or customer-service systems.

C. Stakeholder Analysis

Stakeholder Overview

ParcelResolve touches both business and technical stakeholders, and their expectations are not always aligned. Some care mostly about speed, others about accuracy or cost, and others about technical reliability. Understanding these differences early is important because they will shape requirements that are genuinely in tension with each other.

ST-001: Host / Customer Organization

Role / Interest:

This is the organization that decides to use ParcelResolve and integrates it into its operations. It has the most at stake in terms of overall business outcomes and is the one that will feel the benefit or the cost most directly.

Goals:

- lower the cost and effort involved in resolving incidents;
- get incidents resolved faster and more consistently;
- improve the experience for customers affected by delivery problems;

- cut down on repetitive manual coordination work;
- have better visibility into what types of problems are recurring; and
- connect incident resolution more effectively with existing operations.

Concerns:

- automated decisions that turn out to be wrong;
- the cost of building and running the system;
- whether staff will actually adopt it;
- disruption to processes that currently work well enough;
- data security and protection of personal data; and
- ending up with a system that is more complex than the problem it was meant to solve.

Influence: High**ST-002: Product Owner****Role / Interest:**

The Product Owner is responsible for making sure ParcelResolve is built in the right direction and that the work being done actually creates value. This means making tough calls about scope and priority.

Goals:

- make sure the product addresses a real and meaningful operational problem;
- focus development on the capabilities that matter most;
- keep the scope manageable so the team can deliver something useful;
- support adoption within the organization; and
- show measurable improvement in how incidents are handled.

Concerns:

- scope creep pulling the product in too many directions;
- rising development costs;
- low uptake from operational staff;
- automation that does not justify its complexity; and
- building features that look good on paper but do not match how the organization actually works.

Influence: High**ST-003: External Delivery Stakeholders****Role / Interest:**

This stakeholder category covers the external parties directly affected by a parcel incident. The main roles are the sender, recipient, and, where relevant, a third-party logistics or delivery service provider.

The sender and recipient have different information and may have different expectations:

- **Sender:** The party that originally dispatched the parcel. They may have proof of dispatch, declared value, and information about what was sent. Their concerns may include whether the parcel was handled correctly and whether compensation or another remedy is appropriate.
- **Recipient:** The intended receiver. They are often the person most directly affected by a failed, delayed, or disputed delivery. Their main concern is usually receiving the parcel or getting a clear outcome when that is no longer possible.

Both roles share the same basic interest: a quick, fair outcome with clear communication. However,

their different information positions should be considered when requirements for communication, verification, and evidence collection are developed.

Goals:

- get the issue resolved as quickly as possible;
- receive accurate information about what is happening;
- not have to repeat themselves every time they contact support;
- be treated consistently and fairly; and
- understand clearly what the outcome is and why.

Concerns:

- resolution taking too long;
- being asked for the same information repeatedly;
- the incident being classified incorrectly from the start;
- automated responses that do not fit the actual situation; and
- poor or unclear communication throughout the process.

Influence: Medium**ST-004: Operational User****Role / Interest:**

Operational users are the internal staff who actually work within ParcelResolve on a daily basis — customer service agents, warehouse staff, delivery operations teams, and anyone else who needs to take action on or investigate an incident. They are the ones who will feel most directly whether the system helps or gets in the way.

Goals:

- know clearly what they need to do and when;
- spend less time on repetitive coordination tasks;
- trust that the information the system shows them is accurate;
- have a clear view of which incidents are overdue or escalated; and
- be able to handle unusual situations without having to work around the system.

Concerns:

- automation making the wrong call in edge cases;
- information from integrated systems being incomplete or unreliable;
- responsibility for actions being unclear;
- the system adding to their workload instead of reducing it; and
- not being able to handle real situations that the system was not designed for.

Influence: High**ST-005: IT / Integration Team****Role / Interest:**

This is the team responsible for building, connecting, running, and maintaining ParcelResolve from a technical standpoint. They are the ones who have to make the integrations work in practice, keep the system available, and deal with anything that breaks.

Goals:

- build integrations with existing systems that are reliable and maintainable;
- keep the system available and stable in production;

- design an architecture that can be understood and maintained by future team members;
- protect customer and organizational data; and
- avoid accumulating unnecessary technical complexity.

Concerns:

- how difficult and fragile the integrations turn out to be;
- poor data quality coming from existing systems;
- changes in external systems breaking existing integrations;
- maintenance burden growing over time;
- security vulnerabilities introduced through data exchange;
- unclear boundaries between what ParcelResolve owns and what the integrated systems own; and
- data-protection requirements, including GDPR, given that the system handles personal data such as customer names, delivery addresses, and delivery history.

GDPR and other data-protection obligations are an organizational responsibility. The IT / Integration Team is an important stakeholder in implementing the technical controls needed to meet those requirements.

Influence: High

Stakeholder Conflicts and Tensions

Several tensions emerged from looking at these stakeholders together. They are worth naming explicitly because they will come up during requirements work.

- **Customer vs. Host Organization:** Customers want the fastest possible resolution, but the organization may need verification steps or cost controls before certain actions can be approved.
- **Automation vs. Operational User:** Increased automation is good for the organization, but operational staff need to be able to trust the system's decisions — and have a way to override them when the situation does not fit the standard pattern.
- **Product Owner vs. Host Organization:** Different parts of the organization will want different things from ParcelResolve. The Product Owner's job is to keep the scope focused, which will inevitably mean saying no to some requests.
- **Host Organization vs. IT / Integration Team:** The business side will often want things delivered quickly. The technical team also needs to think about how easy the result will be to maintain and operate, which sometimes means pushing back on the timeline.
- **Customer vs. Operational User:** A customer might want an immediate answer or resolution, while the operational user handling the case may genuinely need to investigate before they can give one.

Stakeholder Priority

For the initial development phase, the highest-priority stakeholders are:

- Host / Customer Organization;
- Product Owner;
- Operational User; and
- External Delivery Stakeholders.

The IT / Integration Team is treated as a critical supporting stakeholder. Their input matters particularly early on, since ParcelResolve depends entirely on reliable data from systems that the IT team owns and maintains.

Key Stakeholder Insight

One thing that becomes clear from this analysis is that ParcelResolve cannot simply optimize for speed. Speed matters, but so do accuracy, fairness, accountability, operational effort, automation quality, data security, and the ability to maintain the system over time.

Many of the requirements tensions that will come up in later stages trace back to this. A decision that improves resolution speed might increase operational user workload or introduce automation risk. A decision that protects customer fairness might slow down the resolution timeline. These are genuine trade-offs, not problems that can be designed away.

D. Tool-Supported Analysis

Tool Used and Interaction

The AI tool used for this section was **Claude** (claude.ai, developed by Anthropic). After completing the initial product vision, scope, and stakeholder analysis, we provided the full draft to Claude and asked it to:

- look for missing stakeholders or gaps in the stakeholder analysis;
- identify scope gaps or underspecified areas;
- flag incorrect assumptions or things that might cause problems in later requirements work; and
- suggest anything else it considered relevant for an incident-resolution system of this kind.

Significant Tool Observations and Suggestions

The tool returned several observations. We assigned each a reference identifier to make it easier to discuss them in Section E.

[T-01] Stakeholder granularity: Sender versus Recipient

Claude pointed out that ST-003 grouped parties together whose interests and available information are meaningfully different. The sender knows what was shipped and may have proof of dispatch, while the recipient is usually the person directly affected by the delivery outcome. The tool suggested treating them as separate stakeholders because their communication and verification requirements could differ.

[T-02] Scope gap: Audit trail and evidence management

For delivery disputes specifically, Claude noted that our scope did not explicitly mention evidence capture. To resolve a dispute consistently, there needs to be access to documented evidence — for example delivery photos, GPS data, signature records, and system timestamps. Without this, dispute resolution could become inconsistent or difficult to justify.

[T-03] Scope suggestion: Compensation and remediation workflow

Claude suggested adding a compensation or refund-initiation capability, on the grounds that many parcel incidents ultimately result in some form of compensation. It argued that a system that tracks incidents to resolution could also trigger this next step.

[T-04] Regulatory consideration: Data protection

The tool observed that no regulatory or compliance stakeholder appeared explicitly in our analysis. It noted that ParcelResolve would handle personal data such as customer names, addresses, and delivery history, making data protection and GDPR relevant.

[T-05] Missing measurable success criteria

Claude noted that the product vision does not include measurable targets, such as a target mean time to resolution or a first-contact resolution rate. Without targets, it would be harder to evaluate whether ParcelResolve has achieved its intended results.

[T-06] Potential resolution gap from the damaged-parcel exclusion

The tool questioned whether excluding damaged-parcel handling could leave a gap. It noted that damage might sometimes be discovered during what started as a lost or delayed parcel case, and asked whether ParcelResolve should define how such mixed incidents are handled.

E. Comparison and Revision**Overview**

We went through each tool suggestion and compared it against our initial analysis. For each one, we decided whether it identified something genuinely missing, something we had already considered but not made explicit, something that is valid but not yet actionable, or something we disagree with.

Assessment of Tool Suggestions**[T-01] Sender versus Recipient distinction — Modified**

The observation is useful. Senders and recipients do hold different information and can have different priorities when a parcel incident occurs. The sender may have information about dispatch and shipment contents, while the recipient is usually closer to the actual delivery experience.

However, splitting them into two completely separate stakeholder entries felt premature at this stage. Both roles ultimately share the same high-level interest — a fair and clear resolution — and separating them now would add complexity without much benefit.

Revision: We revised ST-003 to describe the sender and recipient as distinct roles within one external stakeholder category. Their different information positions will be considered when defining communication, verification, and evidence requirements later.

[T-02] Audit trail and evidence management — Accepted

This was a genuine gap. Delivery disputes are included in our initial incident coverage, but our scope description did not explicitly mention the need to capture and retrieve supporting evidence. Once delivery disputes are included, evidence becomes important for making the resolution process consistent and defensible.

Revision: We added *Evidence capture and retrieval* as an explicit in-scope capability in Section B.

[T-03] Compensation and remediation workflow — Deferred

The suggestion is understandable. Many incidents may eventually lead to compensation or another form of remediation. However, initiating compensation or refunds involves financial authorization, payment systems, and approval workflows that are outside the current product boundary. Payment processing is already listed as out of scope.

We therefore decided not to expand the product scope around this suggestion yet. It can be revisited once the organization's financial and compensation processes are better understood.

[T-04] Data protection and GDPR — Modified

The tool is right that GDPR is relevant here. ParcelResolve will handle personal data, so data protection needs to be considered from the requirements stage onward.

We did not add a separate compliance stakeholder, because the current stakeholder model is focused on the main roles directly involved in operating and developing the product. Instead, we made data protection an explicit concern for both the Host Organization and the IT / Integration Team, while recognizing that overall compliance is an organizational responsibility.

Revision: GDPR and data-protection concerns were made explicit in the stakeholder analysis, with the IT / Integration Team identified as an important technical stakeholder rather than the sole owner of the responsibility.

[T-05] Measurable success criteria — Deferred

The point is fair from a requirements-engineering standpoint. Measurable targets would make it easier to evaluate whether the product is actually improving incident resolution.

However, targets such as resolution time or first-contact resolution should ideally be based on the organization's existing operational data. We do not have those baselines yet.

Decision: We deferred measurable success criteria to a later stage when goals, baselines, and priorities can be defined with more information.

[T-06] Damaged-parcel resolution gap — Rejected

We deliberately excluded damaged-parcel handling from the initial scope, and after reviewing the suggestion, we decided to keep that boundary.

The concern about damage being discovered during another incident is valid, but we do not think it requires adding damaged-parcel handling to the initial product. Instead, such cases can be identified as exceptions and escalated to a human operational user.

The exact handling of mixed incidents involving damage can be defined later when the incident taxonomy and escalation rules are developed in more detail.

Summary of Revisions Made

The tool interaction led to several concrete changes to the initial analysis:

- **ST-003 revised** in Section C to describe sender and recipient as distinct roles within one external stakeholder category.
- **Evidence capture and retrieval** added as an explicit in-scope capability in Section B.
- **GDPR and data protection** made explicit in the stakeholder analysis, while keeping responsibility at the organizational level rather than assigning it only to IT.
- **Incident sources clarified** so that there are two main sources: system-initiated and customer-initiated. Customer service can submit a customer-initiated incident on the customer's behalf.
- **Product lifecycle clarified** as Detect → Understand → Act → Monitor → Verify Resolution → Close.

The compensation suggestion and measurable success criteria were deferred to later requirements work. The damaged-parcel suggestion was rejected because damaged-parcel handling remains outside the initial scope, although mixed incidents involving damage are recognized as a future requirements issue.

F. Tool Reflection

Field	Response
1. Tool / AI Used	Claude (claude.ai, Anthropic)
2. Tool Output / Suggestion	Claude suggested splitting ST-003 into two separate stakeholders: the <i>sender</i> , who has information about the shipment and proof of dispatch, and the <i>recipient</i> , who is usually closer to the actual delivery circumstances. The suggestion was that the two roles could have different communication and verification needs.
3. Decision Taken	Modify
4. Primary Reason Category	Stakeholder need / Requirement quality
5. Justification	The distinction Claude raised is useful because senders and recipients do have different information and may need different things from the resolution process. This could become important when defining detailed requirements for communication, evidence, and verification. However, separating them into two full stakeholder entries at this stage would make the stakeholder model more complicated than it needs to be. At the current level of analysis, both roles share the same basic expectation of a fair, clear, and timely resolution. We therefore modified the original stakeholder instead of fully splitting it. ST-003 now explicitly describes the sender and recipient as different roles and records why the distinction may matter later. This keeps the useful part of the tool's suggestion without adding unnecessary structure to the current analysis.

AI Assistance Disclosure

AI assistance (Claude, claude.ai) was used to help organise and review the product vision, scope, and stakeholder analysis, particularly in Sections D, E, and F. The project definition, scope decisions, stakeholder analysis, and final decisions were discussed and reviewed by the group. Tool suggestions were not accepted automatically; they were considered against the current project scope and either accepted, modified, deferred, or rejected based on our own reasoning.